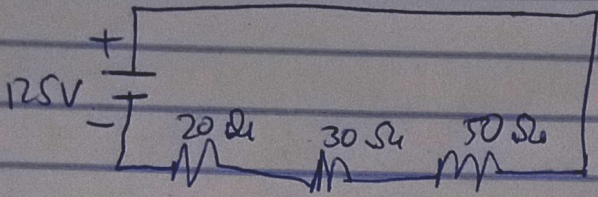


ES10T Assignment 2.

a)



1. Total resistance.

$$\begin{aligned} a) R_t &= R_1 + R_2 + R_3 \\ &= 20 + 30 + 50 \\ &= 100 \Omega \end{aligned}$$

2. Total current.

$$\begin{aligned} I_t &= V / R_t = \frac{125}{100} \\ &= 1.25 \text{ A} \end{aligned}$$

3. Current through each resistor

$$= 1.25 \text{ A} - \text{the same through all three}$$

4. Voltage drop across each resistor.

$$V = (1.25 \times 20) + (1.25 \times 30) + (1.25 \times 50)$$

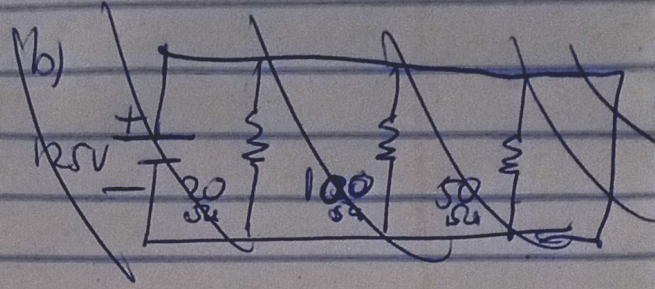
$$V = 25 + 37.5 + 62.5$$

$$V_t = 125 \text{ V}$$

$$= 20 \Omega = 25 \text{ V}$$

$$= 30 \Omega = 37.5 \text{ V}$$

$$= 50 \Omega = 62.5 \text{ V}$$



5. Power dissipated each resistor

$$P = IV$$

$$\begin{aligned} P(R_1) &= 1.25 \times 25 \\ &= 31.25 \text{ W} \end{aligned}$$

$$\begin{aligned} P(R_2) &= 1.25 \times 37.5 \\ &= 46.875 \text{ W} \end{aligned}$$

$$\begin{aligned} P(R_3) &= 1.25 \times 62.5 \\ &= 78.125 \text{ W} \end{aligned}$$

